

# Outer barrier

Wrapped around your body like a protective overcoat, your skin forms a barrier between the inside of your body and the outside world. Your skin is waterproof, keeps out germs, and repairs itself. It also filters out harmful rays in sunlight, enables you to feel your surroundings, and helps control your body temperature.

## UNDER THE SKIN

The largest organ in your body, skin is just millimeters thick. It has two layers—the epidermis and dermis. Cells in the base of the epidermis divide constantly, producing cells that move to the surface, flatten, and fill with a tough, waterproof protein called keratin. The thicker dermis contains sweat glands, blood vessels, and other structures that perform the skin's many tasks.

*The hair shaft sprouts above the skin's surface from a follicle.*

*The epidermis is the upper part of the skin. It has several layers.*

*The dermis is the lower part of the skin. It contains blood vessels.*

*Sebaceous glands release an oily substance, sebum, that softens hair and skin.*

*Arteries supply food and oxygen to skin cells.*

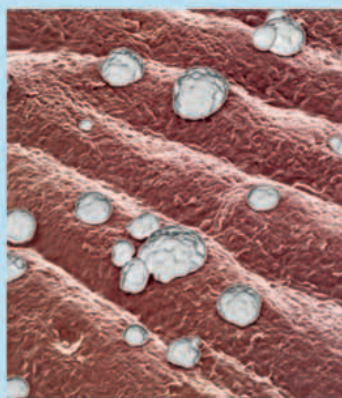
*Nerve carry signals from touch receptors toward the brain.*

**WOW!**

We lose about 50,000 skin flakes every minute, and 44 lbs (20 kg) of flakes in a lifetime.

## LOOK CLOSER: TEMPERATURE CONTROL

Skin performs a vital role in keeping your body temperature at a steady 98.6 °F (37 °C). In hot conditions, dermis blood vessels widen to lose extra heat, and sweat evaporates from the skin to cool the body. When it's cold, dermis blood vessels narrow to keep heat from escaping and tiny muscles pull body hairs upright in an effort to trap warm air, lifting the skin into goosebumps.



▲ **SWEAT** This magnified view shows tiny sweat droplets on the surface of a fingertip.



▲ **GOOSEBUMPS** The skin's surface is covered in little bumps if we get too cold.